

Data Quality Sentinel - Automated Data Validation Framework

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Description

Automated data validation framework. YAML expectations, 12+ check types, statistical anomaly detection, HTML reports with trend tracking.

Tech Stack

pandas, pydantic, pyyaml, jinja2, scipy, click, pytest

Features

- YAML validation expectations
- 12+ check types
- KS-test anomaly detection
- HTML quality reports
- Trend tracking
- CI/CD exit codes

Sample Input

CSV data and expectations

```
sentinel validate --data orders.csv --expectations rules.yaml
```

Sample Output

Quality report

```
Validation: 12/14 checks passed (score: 85.7%)
```

Project Files

```
.gitignore
project-idea.json
pyproject.toml
README.md
.github/workflows/test.yml
sample_data/customers_reference.csv
sample_data/orders.csv
sample_expectations/orders_expectations.yaml
src/data_quality_sentinel.egg-info/dependency_links.txt
src/data_quality_sentinel.egg-info/entry_points.txt
```

```
src/data_quality_sentinel.egg-info/PKG-INFO
src/data_quality_sentinel.egg-info/requires.txt
src/data_quality_sentinel.egg-info/SOURCES.txt
src/data_quality_sentinel.egg-info/top_level.txt
src/sentinel/anomaly.py
src/sentinel/cli.py
src/sentinel/engine.py
src/sentinel/expectations.py
src/sentinel/generate_data.py
src/sentinel/models.py
src/sentinel/report.py
src/sentinel/trend.py
src/sentinel/__init__.py
src/sentinel/checks/builtin.py
src/sentinel/checks/registry.py
src/sentinel/checks/__init__.py
templates/report.html
tests/conftest.py
tests/test_anomaly.py
tests/test_checks.py
tests/test_engine.py
tests/test_expectations.py
tests/test_report.py
tests/test_trend.py
tests/__init__.py
```

Test Results

PASSED

Run: 51 | Passed: 51 | Failed: 0